NIST prime-curve trapdoor slots: a preregistered experimental portfolio (registered 2026-10-11)

Status. Protocol with one executed round (the census, §3) and seven pending experiments (§4–§11). No weakness in any NIST curve is claimed or found. Everything in this file is one of: *executed* (with a command and a hash), *derived* (a formula or a published theorem, cited), *proposed* (a hypothesis with a falsifier), or *not publicly testable*.

Question. If a covert ECDLP trapdoor had been inserted into a NIST prime curve (P-192 … P-521) so that one party could solve ECDLP in sub-exponential time while the public could neither solve it nor recover the trapdoor cheaper than solving it, *where in the published parameters would the trapdoor have to live, and what experiment, run on public data, would bound or exclude each location?*

What this adds to `RESEARCH_P256_BACKDOOR_SPECULATION.md`. That note ranks four hypotheses for P-256. This protocol (i) enumerates every parameter slot of the X9.62 / FIPS 186-4 generation procedure rather than only the curve equation, (ii) runs the public-parameter checks on all five prime curves rather than P-256 alone, and (iii) attaches a registered experiment, with budget and falsifier, to each slot. It reuses the repository's existing probes wherever one exists and names the gap where one does not.

Companion file: `README.md` (reading order and evidence index); `census.md` / `census.json` (executed round 0); `slots.svg` (diagram).

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0. The organising principle: invariants cannot hide, models can

Abstractly `E(F_p)` of prime order `n` is `Z/n`. ECDLP is hard only because of the *representation*. A trapdoor is therefore, by definition, a withheld efficiently-computable homomorphism from the published representation into one with a sub-exponential DLP. Two consequences fix the shape of every experiment below.

1. **Isogeny invariants are public and checkable.** `#E`, the trace `t`, the embedding degree `k = ord_n(p)`, the Frobenius discriminant `t² − 4p` and its square-free part, and hence the CM field, are constant on the whole `F_p`-isogeny class. Any weakness that is a function of these (anomalous, MOV/Frey–Rück, small class group) is excluded by a public check and cannot be hidden by an isogeny walk.
2. **Model-dependent properties can be hidden.** A weakness that depends on the specific Weierstrass model or on the field of definition (the GHS magic number, a specific cover, a specific factor-base structure) is *not* isogeny-invariant. The designer starts at a weak model E_0, walks a secret isogeny chain to `E`, and publishes `E`. Recovering the chain from `E` alone, with the target class unknown, is the asymmetry the trapdoor needs. This is Teske's construction (J. Cryptology 2006) over `F_{2^n}`, `3 | n`, and it is the only published instance.

So the only slot that can hold a sub-exponential ECDLP trapdoor is the representation itself (S1). Every other slot is either a public invariant (checkable), a protocol-level trapdoor that does not touch ECDLP (S4), an implementation-level weakness (S5), or non-weakening (S7). The experiments are arranged so that the checkable slots are *executed* and the representation slot is *bounded* at toy scale by registered, falsifiable predictions.

1. Slot taxonomy

slot	parameter	kind of trapdoor possible	isogeny-invariant?	status in this protocol
S1	the group representation `(p, a, b)` up to isogeny	hidden homomorphism to a sub-exponential group	**no** (that is the point)	S1a–S1d registered, pending
S2	the field prime `p`	special-prime analogue of hidden-SNFS	n/a (field, not curve)	census executed; S2 pending
S3	the seed → `b` map, including the root sign	selection freedom for S1	n/a	census executed (new fact); S3 pending
S4	the base point `G`	protocol trapdoor (Dual_EC class), not ECDLP	n/a	census executed; S4 pending
S5	the quadratic twist	invalid-curve key leak, not ECDLP	twist order is invariant; its *security* is a cofactor fact	census executed; S5 pending
S6	`n`, cofactor, `n ± 1`	none known	yes	census executed
S7	seed as a fingerprint	non-weakening authorship proof	n/a	census executed (public dictionary only)
S8	a better generic algorithm	capability, not trapdoor	n/a	not publicly testable; see boundary ledger

2. Frozen inputs for every round

- Curves: `CurveParams::p192()`, `p224()`, `p256()`, `p384()`, `p521()` from `src/ecc/curve_zoo.rs` (`a = −3`, FIPS 186-4 Appendix D.1.2).
- Seeds: FIPS 186-4 / SEC 2 v2.0 §2.4 values, pinned in `examples/nist_slot_census.rs::seed_for`.
- Derivation: ANSI X9.62-1998 A.3.3.1 as implemented in `src/cryptanalysis/p256_backdoor_map.rs::x962` / `verify_seed_derivation` (already pinned by a test for P-256).
- Repository revision for round 0: `b5bd97f2d302370be58d2cf31f75bd1490a96c99` plus this PR's files. Toolchain: rustc 1.97.0 (2d8144b78 2026-07-07), 2 × Intel Xeon @ 2.10 GHz, Linux 6.18.
- Trial-division bound 10^6; Pollard rho budget 2^{22} iterations per split; embedding-degree scan $k \le 1000$.

3. Round 0, executed: the public-parameter census

Command.

```
cargo build --release --example nist_slot_census
./target/release/examples/nist_slot_census          > census.md
./target/release/examples/nist_slot_census --json   > census.json
```

Wall time 35 s (the Pollard rho budget on the five twist orders).

Hashes.

```
sha256(examples/nist_slot_census.rs) = c59d84f8071c2bebd512abdeb206ccac6aa6c3165e216e7f8520ef0211
sha256(census.json)                   = 3467a6d68eae2b57f76a3db020ca64794bbccd713462ce9f5e060ec710
sha256(census.md)                     = 9f7419e3668ea20529f6950c5053ebe0580779816689cf0b111a8266e0
```

Observed (from `census.md`; every row recomputed, nothing pinned).

Group invariants (S2, S5, S6). All five curves: prime order, cofactor 1, trace positive, embedding degree > 1000. $|D| = 4p - t^2$ has no square factor from primes below 10^6 except P-224 (3^3 contributes a 4-bit square part); no residue is a perfect square; the P-192 residue is a 184-bit probable prime. So no public small-class-group or small-conductor structure is reachable by trial division on any of the five. $n - 1$ and $n + 1$ carry only small factors below the bound with large residues (P-224's $n - 1$ residue is a 196-bit probable prime).

Seed → b and the root sign (S3). $r(\text{seed}) \cdot b^2 \equiv a^3 \pmod p$ holds for all five published seeds. The procedure fixes b^2, not b; the generator chose a sign. For P-192, P-256, P-384 and P-521, $p \equiv 3 \pmod 4$, so -1 is a non-residue and $b \mapsto -b$ is the *quadratic twist*: the sign choice selects between a curve and its twist after the hash output is known. For P-224, $p \equiv 1 \pmod 4$, $i \in \mathbb{F}_p$, and $b \mapsto -b$ is an $\mathbb{F}_p$-isomorphism $(x, y) \mapsto (-x, iy)$: no freedom. The published root is the numerically smaller one for P-192, P-256 and P-521 and the *larger* one for P-224 and P-384. No published convention explains the choice; this is one extra, previously unremarked bit of generator freedom on four of the five curves, beyond the seed itself. It is not evidence of grinding: it is one bit.

Generator (S4). G is on every curve. $G.x$ is never the lexicographically first valid x (that is $x = 0$ on P-192, P-256, P-384, P-521 and $x = 3$ on P-224), and $G.y$ is the larger of the two roots on P-224 and P-521. So G is not the deterministic "first point" a rigid procedure would produce; its provenance is unexplained on all five curves, as the P-256 literature already notes.

Twist (S5). Twist orders are composite for P-192, P-224, P-256, P-521 and prime for P-384. With the rho budget: P-256's twist is $3 \cdot 5 \cdot 13 \cdot 179 \cdot q_{241}$ (q_{241} a 241-bit probable prime), P-521's is $5 \cdot 7 \cdot 69_697_531 \cdot 635_884_237 \cdot q_{461}$, P-224's is $3^2 \cdot 11 \cdot 47 \cdot 3_015_283 \cdot 40_375_823 \cdot c_{166}$ with a 166-bit composite the budget did not split, and P-192's is $23 \cdot c_{188}$. The P-224 twist is the known weak one (SafeCurves: rho on the twist near 2^58); the unsplit composite is exactly where that figure lives and is S5's pending item.

Seed relations (S7). 84 public nothing-up-my-sleeve candidates per curve (SHA-1 of the curve's names, of p and n in decimal and hex, with and without newline; the other four seeds; SHA-1 of each seed and of each seed + 1 … 4; $\text{name} \parallel \text{counter}$): zero matches on all five. This excludes only those families; a $\text{SHA-1}(\text{secret} \parallel \text{counter})$ seed is undetectable by construction.

What round 0 does and does not establish. It establishes that no public, trial-division-reachable weak class is present on any of the five curves and that the generation procedure leaves two unexplained degrees of freedom per curve besides the seed: the root sign (four curves) and G (all five). It does not bound S1, which is the only slot that can carry a sub-exponential trapdoor; S1 is the subject of the pending rounds.

4. S1a — prime-field cover route (pending)

Hypothesis H1a. There is no cover $C \to E$ with C defined over $\mathbb{F}_p$ (or over $\mathbb{F}_{p^k}$ with E base-changed) whose Jacobian index calculus beats rho on $E(\mathbb{F}_p)$; the complexity exponent $\tilde{O}(p^{2 - 2/g})$ of double-large-prime index calculus on $\text{Jac}(C)$ is above rho's $1/2$ for every genus, and base change to $\mathbb{F}_{p^k}$ only raises it. *Derived* already as Prediction A-1 in research/cover_route_prime_binary_20261008/PROTOCOL.md.

What is not derived. Whether a *specific* small-degree cover can be hidden by an isogeny walk and still transfer the subgroup. Over $\mathbb{F}_p$ the answer is "irrelevant" if H1a holds (the destination solver is never faster), so the experiment is a toy-scale confirmation of the exponent, not a search.

Experiment E1a. For $p \in \{2^{20} + c, 2^{24} + c, 2^{28} + c\}$ (first primes above those powers), 32 random prime-order curves each. For each, construct the degree-2 covers $C: w^2 = f(x)$ with f the cubic pulled back through $x \mapsto x^2$ and $x \mapsto x^2 + u$ (genus 2) and the degree-3 cover through the Kummer line (genus 3 or 4), using src/cryptanalysis/jv_cover.rs and research/hyperelliptic_cover_infrastructure_20261006. Run the repository's hyperelliptic index calculus on $\text{Jac}(C)$ with the standard large-prime variant and rho on E, both in ecbench's counted unit. **Prediction:** measured $\log_p(\text{cost}(\text{Jac}))$ slope across the three p is within 0.1 of $2 - 2/g$ and above $0.5 + 0.05$ for every cover. **Falsified if** any cover family has slope below 0.55. Stop condition: 3 p values × 32 curves × 3 cover families, or 48 CPU-hours.

5. S1b — class-group map (pending)

Hypothesis H1b. No efficiently computable homomorphism `E(F_p) → Cl(O)` exists for `O = End(E)` unless `gcd(n, h(O)) > 1`, because a homomorphism of prime-order `Z/n` into a group of order `h` is trivial when `gcd(n, h) = 1`. *Derived* (Lagrange). For the NIST curves `h(O)` is not computable (`|D|` is 194–523 bits with no public small-conductor structure, census §3), so H1b cannot be tested on them; it can only be tested on the *density* question: how often does a random prime-order curve have `gcd(n, h(O)) > 1`, and could a 1999 grind reach that class?

Experiment E1b. For `p` the first prime above `2^28`, sample 2^{16} random prime-order curves (point count by the repository's SEA or BSGS at that size), compute $D = t^2 - 4p$, its square-free part and `h(D)` by the repository's class-number routine (`hilbert_class_poly.rs` machinery or a baby-step enumeration of reduced forms at 30 bits), and record `gcd(n, h)`. **Prediction:** the fraction with `gcd(n, h) > 1` is below `2^{-12}` and every such case has `h` with a small factor dividing `n` trivially (since `n` is prime, `gcd > 1` means `n | h`, which for $h \approx \sqrt{|D|} \approx p^{1/2} < n$ is impossible). So the registered prediction is **zero** such curves; **falsified if** any curve has `n | h(O)`. This is a sanity bound, recorded so that S1b has a measured row rather than an argument. Cost: one point count and one class-number computation per curve at 30 bits, ≈ 2 CPU-hours.

6. S1c — hidden factor-base structure for summation-polynomial index calculus (pending)

Hypothesis H1c. The point-decomposition cost in Semaev index calculus over `F_p` depends on `p` and the decomposition arity, not on the curve: isogenous curves have equal decomposition success probability and equal Gröbner / SAT solve cost to within noise, so no model in an isogeny class is a "better factor base carrier" than another. *Proposed.*

Experiment E1c. For `p` the first prime above `2^24`, take 8 random prime-order curves and for each walk 16 steps of the repository's non-backtracking `ell`-isogeny walk (`isogeny_walk/`, `p256_isogeny_walk.rs` generalised to small `p`). On each of the 136 vertices, run the frozen Semaev arity-3 decomposition harness (`semaev_decomp.rs`, the boundary-ledger protocol for regime C) with the same factor-base construction (x-coordinate below `p^{1/3}`), 1024 random targets each. Record decomposition success rate and mean solve time in the counted unit. **Prediction:** within each isogeny class the success rate is constant to within binomial `3σ` and the solve time within `10 %`. **Falsified if** any vertex differs from its class mean by more than that on both metrics. A falsification would be the first evidence that a model-dependent factor-base advantage exists, and would be the prime-field analogue of the GHS magic number. Cost: 136 × 1024 decompositions at 24 bits, ≈ 6 CPU-hours.

7. S1d — isogeny-hiding asymmetry (pending)

Hypothesis H1d. (i) Every public weak-class predicate (anomalous, `k ≤ 20`, square-free part of `D` below `2^40`) is constant along the walk; (ii) recovering a hidden chain of `L` steps of degree `ell` to a *known* target costs `O(√L · ell)` by meet-in-the-middle and to an *unknown* target costs no less than enumerating the class, so the asymmetry a Teske-style trapdoor needs exists over `F_p` *for the hiding step*, even though no weak class exists to hide. (i) is *derived* (Tate); (ii) is *proposed* at the measured level.

Experiment E1d. For `p` the first prime above `2^28`, 8 random prime-order curves, walks of `L ∈ {8, 16, 32}` steps at the smallest split `ell`. Assert (i) on every vertex with the repository's invariant checks. For (ii), implement the bidirectional meet-in-the-middle recovery with the target known, and the class-enumeration recovery with only the *predicate* known (here the predicate is "has `j`-invariant in a random 1 % subset", standing in for a secret class), both counting modular-polynomial root extractions. **Prediction:** known-target recovery cost grows as `ell^{L/2}` within a factor 4; predicate-only recovery grows as the class size divided by the subset density, i.e. is independent of `L` and ≈ `100 × √|D|/ell`-ish at this `p`; the ratio between the two exceeds `2^10` at `L = 32`. **Falsified if** predicate-only recovery is within `2^5` of known-target recovery at `L = 32`. Cost: ≈ 4 CPU-hours.

8. S2 — the prime (pending, one new control)

Hypothesis H2. The Solinas shape of `p` confers no advantage to any ECDLP method the repository can run. Covered for summation-polynomial index calculus by `research/notes/cm-isogeny/RESEARCH_NIST_SOLINAS_EXPERIMENTS.md` experiment 4 (measured no gain). `p ± 1` smoothness only reaches `F_p^*`, which `k > 1000` (census) never touches.

Experiment E2. The one control that file did not run: matched rho. Run `p256_negation_rho` (`p256_attacks.rs`) on P-256 and on a random 256-bit prime-order curve over a random (non-Solinas) 256-bit prime with the same ecbench spec, 2^{20} iterations each, and record the per-iteration counted cost. **Prediction:** the Solinas reduction is faster per field multiplication (that is its purpose) and the *iteration count to collision* is statistically identical; the protocol is a stage diagnostic and makes no end-to-end claim. **Falsified if** the Solinas curve needs fewer iterations at `3σ`.

9. S3 — seed grinding density at toy scale (pending)

Hypothesis H3. The X9.62 procedure at toy scale reproduces the grind-budget table of `RESEARCH_P256_BACKDOOR_SPECULATION.md` §3: the density of prime-order outputs is `≈ 1/ln p`, and the density of each public weak class among prime-order outputs is as the literature gives (anomalous `≈ 1/√p`-ish per Hasse interval, `k ≤ 20` `≈ 20/p`, square-free part of `D` below `B` `≈ B/√p`).

Experiment E3. Implement the full X9.62 A.3.3.1 loop natively for `p` of 40 bits (first prime above `2^40`), `a = −3`, 2^{24} sequential seeds. For each candidate compute `b` (both roots), the curve order (the repository's point counter at 40 bits), and the public weak-class predicates. Record densities with Wilson intervals. **Prediction:** each density within a factor 2 of the formula. **Falsified if** any public class is denser than `2^{−16}` among prime-order outputs. A second output of E3 is the *root-sign* statistic: whether the smaller root yields a prime-order curve more or less often than the larger (prediction: no difference at `3σ`). Cost: 2^{24} point counts at 40 bits, ≈ 20 CPU-hours; run on isolab per `ecbench-independent-runner`.

10. S4 — base point (pending, two items)

Experiment E4a (rigid-derivation test). Pin the ANSI X9.62-1998 Annex A.3.2 base-point derivation and the SEC 1 / FIPS 186-4 text on `G` (none of which specify a seeded derivation for the NIST `Gs`; this must be confirmed from the standards' text and recorded in the round report). Then test each published `G` against every candidate deterministic rule the round can state exactly: lexicographically first `x` (done, census: no), `x = SHA-1(seed ‖ i)` reduced for `i ≤ 2^16`, `x = SHA-1(seed)` with the X9.62 `r` construction, and `G = [h]·(first point)` with `h = 1`. **Prediction:** no match. A match would close S4 in the mundane direction for that curve.

Experiment E4b (protocol audit, no computation). Enumerate deployed protocols on NIST prime curves that expose scalar multiples of *two* fixed points or of `G` plus a long-term point in a way that makes a secret relation `Q = [d]G` exploitable (the Dual_EC pattern): Dual_EC_DRBG (withdrawn), ECDSA with deterministic nonce derivation (no), ECIES/ECDH (no second fixed point), the ECQV implicit-certificate scheme, and the EC-based KEMs in the ISO 18033 family. Deliverable is a table with the trapdoor condition per protocol. This is a literature round and makes no ECDLP claim.

11. S5 — twist completion (pending)

Experiment E5. Finish the factorisation of the P-192 and P-224 twist orders (the 188-bit and 166-bit unsplit composites of census §3) with ECM in the repository's native factoring tools (or a bounded `cargo` ECM), so the "rho on the twist" cost is a measured number rather than a literature citation. **Prediction:** the P-224 twist's largest prime factor is near `2^117` (SafeCurves `2^58.4` rho cost ×2) and P-192's is above `2^150`. Then re-run the invalid-curve harness (legacy Python in the repository; per `AGENTS.md` it must be replaced by a native path before being extended, so this round includes that replacement) against a test ECDH endpoint that skips point validation, and record the

per-query leak in bits, which the census already bounds from below by the small factors (15.1 bits for P-256, 12.2 for P-224). This is an implementation-level result and is reported as such.

12. S6, S7, S8 — closed or not testable

- **S6.** Executed in census §3. No pending item: n prime, cofactor 1, no known consequence of n ± 1 smoothness.
- **S7.** The public dictionary is executed (zero matches). The remaining family, SHA-1(secret ‖ counter), is undetectable; the only closing evidence is a published preimage (the 2023 bounty). No experiment can be registered.
- **S8.** A classified generic improvement is not a parameter trapdoor and no public parameter check bears on it. The repository's evidence on this is the boundary ledger (docs/ic/BOUNDARY_TARGETS.md, regime C), which records the measured prime-field index-calculus frontier against rho. Nothing is registered here.

13. Claim hygiene and what success would mean

Per AGENTS.md: every round reports in the counted unit against the ledger, fail-closed, synthetic fixtures only; no key recovery, no sub-rho asymptotic claim, no deployed-curve-security claim. The portfolio's possible outcomes are:

- all predictions hold: the slots S2–S7 are *measured* clean on all five curves and S1 is *bounded* at toy scale; the residual is S8 and the undetectable S7 family;
- a falsification in E1c or E1d: a model-dependent property that survives isogeny hiding over F_p exists at toy scale and becomes a research lead for the index-calculus ladder, reported as a stage diagnostic with the exact curve identities;
- a falsification in E3: the X9.62 procedure at toy scale is biased in a way the literature formulas do not predict, which would re-open the grind budget of RESEARCH_P256_BACKDOOR_SPECULATION.md §3.

None of these outcomes would by itself establish that a NIST curve is or is not trapdoored. The portfolio bounds *where* a trapdoor could be and *what it would cost to have placed one*, which is the most a public check can do.

14. References

- ANSI X9.62-1998, Annex A.3.3; FIPS 186-4, Appendix D.1.2; SEC 2 v2.0 §2.4.
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- D. J. Bernstein et al., *How to manipulate curve standards*, SSR 2015.
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- C. Diem, *On the discrete logarithm problem in elliptic curves*, Compositio 2011.
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research/p256_isogeny_walk_20261004/PROTOCOL.md, src/cryptanalysis/ec_trapdoor.rs.

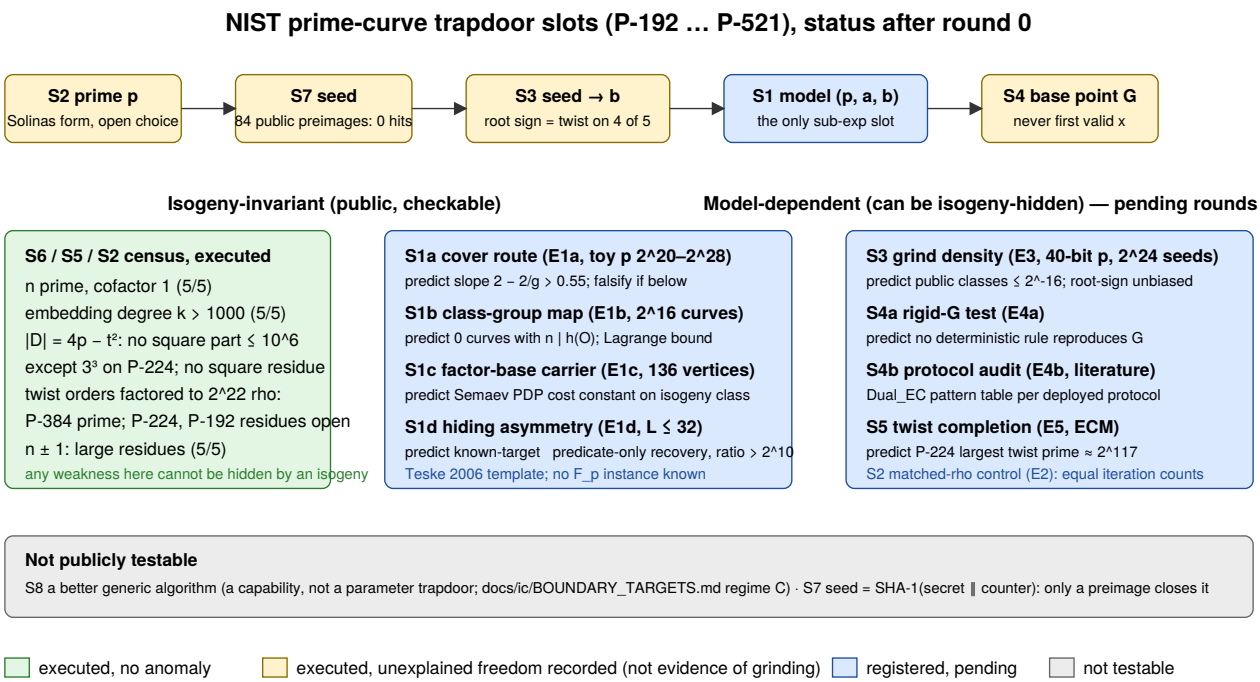


Figure 1: Slot taxonomy and round-0 status

Diagram

Round 0 census (verbatim)

NIST prime-curve trapdoor-slot census

Executed by `examples/nist_slot_census.rs`. Every value is recomputed from `CurveParams` and the published seeds; nothing is pinned by hand.

Group invariants (slots S2, S5, S6)

curve	p bits	trace sign	n prime	k ≤ bound	|D| bits	square part of |D| from small factors (bits)	|D| residue (bits)	residue square?	residue prime?
P-192	192	+	true	> 1000	194	1	184	false	true
P-224	224	+	true	> 1000	226	4	182	false	false
P-256	256	+	true	> 1000	258	1	255	false	false
P-384	384	+	true	> 1000	386	1	386	false	false
P-521	521	+	true	> 1000	523	1	521	false	false

X9.62 root-sign freedom and twist (slots S3, S5)

curve	seed → b holds	p mod 4	sign choice = twist	published root smaller	twist order prime	twist leak bits (< trial bound)	twist residue bits	residue prime
P-192	true	3	true	true	false	4.5	188	false
P-224	true	1	false	false	false	12.2	212	false
P-256	true	3	true	true	false	15.1	241	true
P-384	true	3	true	false	true	0.0	385	true
P-521	true	3	true	true	false	5.1	516	false

Generator provenance (slot S4)

curve	G on curve	smallest valid x	G.x is smallest valid x	G.y is smaller root	x scanned
P-192	true	0x0	false	true	1
P-224	true	0x3	false	false	4
P-256	true	0x0	false	true	1
P-384	true	0x0	false	true	1
P-521	true	0x0	false	false	1

Twist order factorisation (trial division + bounded Pollard rho)

curve	factors (hex, exponent, probable prime)	largest proved prime factor (bits)
P-192	0x17^1 (prime) · 0xb21642c8590b21642c8590b25d4ea13d29723c09f184109 (unsplit residue, composite)	5
P-224	0x3^2 (prime) · 0xb^1 (prime) · 0x2f^1 (prime) · 0x2e0273^1 (prime) · 0x268160f^1 (prime) · 0x20905fa28a700b3b10f0d8d3df7028d8feb21bc6ff (unsplit residue, composite)	26
P-256	0x3^1 (prime) · 0x5^1 (prime) · 0xd^1 (prime) · 0xb3^1 (prime) · 0x1e0a75640070a738557cc30f68bd56eaea65c94f98411d17ac4e16ece1a47^1 (prime)	241
P-384	0x1000000000000000000000000000000000000000000000000000003895cb27e0bc8d (prime)	385
P-521	0x5^1 (prime) · 0x7^1 (prime) · 0x4277ffb^1 (prime) · 0x25e6d2cd^1 (prime) · 0x17c8b8fa594c0fc63a5c0043ab498c1762d92f18fdfe2fea8f074695615d886d8 (prime)	461

|D| small factors

- P-192: [(5, 1), (11, 1), (31, 1)]
- P-224: [(3, 3), (29, 1), (79, 1), (7523, 1), (40927, 1)]
- P-256: [(3, 1), (5, 1)]
- P-384: []
- P-521: [(5, 1)]

n ± 1 smoothness (slot S6)

curve	n−1 small factors	n−1 residue bits	n−1 residue prime	n+1 small factors	n+1 residue bits	n+1 residue prime
P-192	[(2, 4), (5, 1), (2389, 1)]	175	false	[(2, 1), (3, 1), (41, 1)]	185	false
P-224	[(2, 2), (3, 6), (5, 1), (17, 1), (2153, 1)]	196	true	[(2, 1), (7, 1), (19, 1)]	216	false
P-256	[(2, 4), (3, 1), (71, 1), (131, 1), (373, 1), (3407, 1), (17449, 1), (38189, 1)]	188	false	[(2, 1), (5, 1), (1879, 1)]	242	false

curve	n−1 small factors	n−1 residue bits	residue prime	n+1 small factors	n+1 residue bits	residue prime
P-384	[(2, 1), (3, 2), (7, 2), (13, 1)]	371	false	[(2, 2), (83, 1), (181, 1), (421, 1)]	360	false
P-521	[(2, 3), (7, 1), (11, 1), (1283, 1)]	502	false	[(2, 1), (3, 2), (5, 2)]	513	false

Seed relations (slot S7)

- P-192: 84 candidates checked, 0 matched
- P-224: 84 candidates checked, 0 matched
- P-256: 84 candidates checked, 0 matched
- P-384: 84 candidates checked, 0 matched
- P-521: 84 candidates checked, 0 matched